

Methods Workshop

ECON 692 – Applied Economics Seminar

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Today's Agenda

Time	Duration	Activity
4:35	20 min	Check-in & strategy draft debrief
4:55	30 min	Stress-test your design
5:25	40 min	Event studies: theory & code
6:05	10 min	Break
6:15	25 min	Claude Code for estimation
6:40	50 min	Work sprint
7:30	15 min	Share-out & spring break planning

Check-in

How did the empirical strategy draft go?

Quick round: one sentence on where you stand.

Today's Goal

Make **meaningful progress on your empirical results** — run your first regression or refine what you have.

Learning Objectives

1. **Diagnose** threats to your identification strategy
2. **Specify and interpret** an event study in fixest
3. **Use Claude Code** as a pair programmer for estimation
4. **Run your main specification** on your data

Strategy Draft Debrief

What I Saw in Your Drafts: Strengths

What's working:

- Clear **DAGs** mapping causal relationships
- Well-specified **estimating equations**
- Identified **threats** to your designs
- **Novel data sources** (satellite imagery, LLM-scored text, proprietary platforms)
- **Modern methods** (SDID, BSTS, event studies)

What I Saw in Your Drafts: Common Gaps

Patterns to watch out for:

1. **Too long** – most drafts exceeded the page limit. Be concise; every sentence should earn its place
2. **DAGs missing the treatment variable** – your DAG must show the path from *treatment* to *outcome*; without it, you cannot identify confounders
3. **DAG and identification strategy are disconnected** – your strategy section should explain *which* paths in the DAG are problematic and *how* your method addresses them

What I Saw in Your Drafts: Common Gaps (cont.)

4. **Treatment variation is unclear** – “What changes, for whom, and when?” needs a crisp answer
5. **Pre-trends not tested** – plans mention parallel trends but don’t specify *how* to test them
6. **FE vs. treatment collinearity** – if treatment only varies over time, year FE absorb it
7. **Robustness plan is vague** – “I will run robustness checks” is not a plan; name specific tests

Quick Fixes: Tools for Better Drafts

Equations: Use $\text{\LaTeX}$ — either in a .tex document or via the **Auto- $\text{\LaTeX}$ Equations** plugin for Google Docs. $Y = a + bX + e$ is hard to read; typeset equations are not.

DAGs: Use **ggdag** or **DAGitty** (dagitty.net). Both produce cleaner DAGs than drawing by hand and can identify adjustment sets for you.

Minimum Standard

Your empirical strategy draft should have **typeset equations** and a **programmatically generated DAG**. Fix these before submitting preliminary results.

Stress-Test Your Design

The Five Questions

Every empirical design must answer these **five questions**. If you can't, your referee will.

1. **What is your source of variation?** — What changes, for whom, and when?
2. **What is your identifying assumption?** — State it in *words*, not just math.
3. **What would violate it?** — Name two specific threats.
4. **How would you know if it's violated?** — What test or plot would show the problem?
5. **What is your strongest robustness check?** — Placebo, falsification, alternative spec, or bounding exercise?

Example: DiD with Parallel Trends

“I use a difference-in-differences design comparing treated and untreated counties around a policy change.”

1. **Source of variation:** Policy adopted in some counties in 2018, not others
2. **Identifying assumption:** Absent the policy, treated and untreated counties would have followed the same trend
3. **What violates it:** (a) Differential pre-trends; (b) Correlated policy adoption (counties with rising outcomes adopt the policy)
4. **How to detect:** Event-study plot — pre-period coefficients should be near zero
5. **Strongest robustness:** Placebo treatment date; drop early adopters; add county-specific linear trends

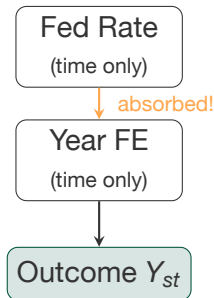
Pitfall: Fixed Effects Absorb Your Treatment

The problem: Your treatment varies **only over time** (e.g., the federal funds rate). You include **year fixed effects**.

Year FE absorb all time-varying common shocks — including your treatment.

The fix:

- Find **cross-sectional variation** in exposure
- Interact treatment with a unit-level characteristic
- Use an **instrument** that varies across units



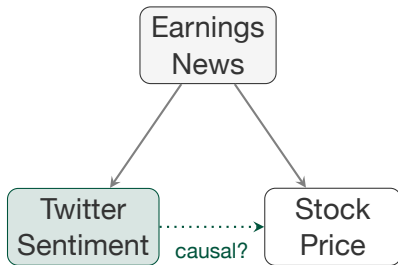
Pitfall: Endogenous “Treatment”

The problem: Your treatment is **endogenous** — it responds to the outcome.

Example: “Does negative sentiment cause stock declines?” But bad earnings *cause* both negative sentiment and price drops.

The fix:

- Find an **exogenous shock** to sentiment
- Use **lagged** sentiment to break simultaneity
- Reframe as a **prediction** question if causation is not feasible



Pitfall: No Control Group $\neq$ No Identification

Some designs don't need a traditional control group.

BSTS / CausalImpact

- Constructs a **synthetic counterfactual** from covariates
- No untreated units needed
- Key: covariates predict outcome well in pre-period

Synthetic Control / SDID

- Builds a **weighted combination** of donor units
- “Control group” is constructed, not observed
- Key: synthetic unit tracks treated unit pre-treatment

Key test: How well does the counterfactual fit the pre-period? Run **placebo tests** — fake treatment dates or fake treated units.

Workshop: Stress-Test Your Design

5 minutes — pair exercise

1. Open your empirical strategy draft
2. Write down answers to all **five questions** for your own project
3. Swap with a neighbor — read their answers and give one piece of critical feedback

Prompts for the Critic

- “Can you state the identifying assumption without using math?”
- “What is the most plausible alternative explanation for your result?”
- “What would a skeptical referee say?”

Event Studies: Theory & Code

Why Event Studies?

An event study does **two things**:

1. **Tests parallel trends** — pre-treatment coefficients near zero $\Rightarrow$ credible comparison group
2. **Shows dynamic effects** — immediate? Growing? Fading?

Every DiD paper should include an event-study plot. Referees expect it.

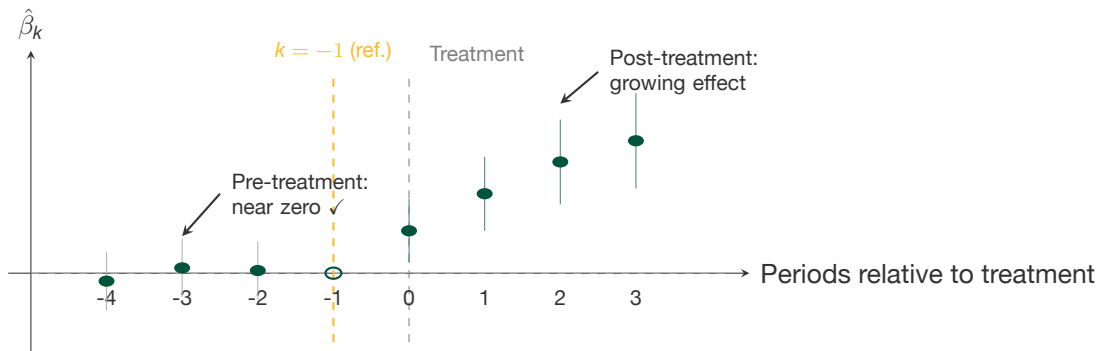
Using panel FE instead of DiD? The same logic applies: plot coefficients relative to a key date.

The Event-Study Specification

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[\text{time-to-treat}_{it} = k] + \varepsilon_{it}$$

- α_i — unit fixed effects (absorb time-invariant differences)
- γ_t — time fixed effects (absorb common shocks)
- β_k — effect at k periods relative to treatment
- $k = -1$ is the **reference period** (omitted, normalized to zero)
- Pre-treatment β_k ($k < -1$): should be ≈ 0 if parallel trends hold
- Post-treatment β_k ($k \geq 0$): the dynamic treatment effect

Anatomy of an Event-Study Plot



Reading Event-Study Plots: What to Look For

Good signs:

- Pre-treatment $\hat{\beta}_k \approx 0$ with CIs crossing zero
- Clear break at treatment date
- Post-treatment coefficients consistent sign
- CIs don't explode at endpoints

Red flags:

- Pre-treatment coefficients trending
- “Effect” appears *before* treatment
- Huge CIs everywhere (underpowered)
- Significant pre-treatment coefficients

Common mistake: “Pre-trends are insignificant, so parallel trends hold.”
Insignificance $\neq$ zero. Look at *magnitudes*, not just *p*-values.

Event Studies in fixest: Basic DiD

```
library(fixest)

df <- read.csv("data/panel_clean.csv")

mod_did <- feols(
  y ~ treat:post | unit_id + year,
  data = df,
  cluster = ~unit_id
)

summary(mod_did)
```

Output: treat:post 0.045** (0.018) — treatment increases Y by 0.045 units.

Event Studies in fixest: The Event Study

```
df$time_to_treat <- df$year - df$treat_year
df$time_to_treat[is.na(df$treat_year)] <- -1000

es <- feols(
  y ~ i(time_to_treat, treated, ref = -1) |
    unit_id + year,
  data = df,
  cluster = ~unit_id
)

iplot(es, main = "Event Study: Effect on Y")
```

That's it. Four lines of code from data to event-study plot.

Customizing the Event-Study Plot

```
ipplot(es,  
  main = "Dynamic Treatment Effects",  
  xlab = "Years Relative to Treatment",  
  ylab = "Coefficient Estimate",  
  col  = "darkgreen",  
  ref.line = -1,  
  ref.line.par = list(col = "gray", lty = 2)  
)  
abline(h = 0, lty = 2, col = "gray60")
```

Save with: `png("output/figures/event_study.png", width=8,
height=5, units="in", res=300)`

What If You Can't Run an Event Study?

Not every design supports a standard event study. Use these instead:

Method	Diagnostic	Package
SC / SDID	Placebo units; pre-fit plot	augsynth, synthdid
BSTS	Placebo dates; covariate balance	CausalImpact
Panel FE	Coefficient stability; Oster bounds	fixest
IV / 2SLS	First-stage F; reduced form	fixest

Same principle: show the reader your variation is credible.

Break

10-minute break

Up next: Claude Code for estimation

Claude Code for Estimation

Claude Code as a Pair Programmer

Claude is good at:

- Turning your strategy into R/Python code
- Debugging data issues (merges, missing values, types)
- Writing `fixest`, `did`, `augsynth` code from a description
- Generating robustness checks
- Formatting tables and figures

You own:

- The identification argument
- Whether your design is valid
- Interpreting coefficients in context
- Which robustness checks matter
- The narrative of your paper

You are the economist. Claude is the coder.

The Key: Your CLAUDE.md

Claude Code reads CLAUDE.md on every prompt. Put your empirical strategy there.

CLAUDE.md -- My Capstone Project

Research Question:

Does CA Paid Family Leave reduce the gender wage gap?

Data:

CPS ASEC, 2000–2012, state–year panel

Treatment: CA adopts PFL **in** 2004

Outcome: female-to-male earnings ratio

Empirical Strategy:

Method: Synthetic DiD (Arkhangelsky+ 2021)

Donors: 44 states (excl NJ, WA, NY, RI, HI)

Pre: 2000–2003; Post: 2004–2012

Package: synthdid **in** R

Key Threats:

Housing bubble **in** CA (2004–2006)

Immigration surge (compositional change)

Demo: From Strategy to First Regression

With a good CLAUDE.md, you can prompt like this:

```
> Write my main SDID specification using synthdid.  
   Use my CPS data and the design in CLAUDE.md.
```

Then iterate:

```
> Now run placebo tests on the donor states.  
> Add a pre-period fit plot comparing CA  
   to the synthetic CA.  
> Save all figures to output/figures/ at 300 dpi.
```

Each prompt builds on the last. CLAUDE.md gives Claude persistent context.

Tips for Effective Prompting

1. **Put your empirical strategy in `CLAUDE.md`**

No need to re-explain your design every time

2. **Name specific packages**

“Use `fixest::feols`” not “run a regression”

3. **Ask for diagnostics alongside estimation**

“Run the model and check for pre-trends” gets both in one pass

4. **Read the code before running it**

You own the analysis. Check what it writes.

5. **Commit after each working version**

`git commit` — you can always roll back

What Not to Do with Claude Code

1. **Don't outsource your thinking**

“Is my identification valid?” — that's *your* job

2. **Don't run code you haven't read**

Understand each line before interpreting results

3. **Don't treat it as an oracle**

Claude can write wrong code. Verify.

4. **Don't skip the economics**

A perfect `fixest` spec with the wrong identifying assumption is still wrong

Rule of thumb: If you can't explain every line of code and every coefficient, you're not ready to present.

Work Sprint

Work Sprint: Start Your Estimation

50 minutes — hands on

Setup (first 5 minutes):

1. Open your project in the terminal
2. Ensure CLAUDE.md includes your empirical strategy
3. Have your data loaded and cleaned (from Lecture 05)

Tasks:

1. Write your **main specification** in R (or Python)
2. Run it and **interpret the coefficient** — one sentence
3. Add one **diagnostic**: event study, placebo, balance table, or pre-fit plot

Raise your hand if stuck. I will circulate.

Sprint: Method-Specific Starting Points

Your Method	First Command	Then Ask Claude
DiD / Event Study SDID	<code>feols(y ~ treat:post id + t)</code> <code>synthdid_estimate(Y, N0, T0)</code>	“Add an event study” “Run placebo tests”
BSTS	<code>CausalImpact(data, pre, post)</code>	“Plot the counterfactual”
Panel FE	<code>feols(y ~ x id + t)</code>	“Add controls one at a time”
IV / 2SLS	<code>feols(y ~ 1 id + t x ~ z)</code>	“Report first-stage F”

If you get stuck on data issues, ask Claude: “My data has [problem]. How do I fix it before running the regression?”

Sprint Checkpoint

Before you stop, verify:

- ☐ I ran my main specification and got a coefficient
- ☐ I can state in one sentence what the coefficient means
- ☐ I started at least one diagnostic (event study, placebo, balance)
- ☐ My code is saved and version-controlled (`git add + git commit`)

All four? You're ahead of schedule for **Preliminary Results** (due March 27).

Share-Out & Wrap-Up

Share-Out

3–4 volunteers, 3 minutes each

Come up to the front and tell us:

1. What specification did you run?
2. What is the coefficient, and what does it mean?
3. What surprised you?

Spring Break Homework

Preliminary Results are due **Friday, March 27** (day after first class back).

Minimum deliverable:

- Main specification estimated and interpreted
- At least one robustness check
- Code is reproducible

Stretch goals:

- Event-study plot (if applicable)
- Multiple specs, coefficient stability
- Draft results section (2–3 paragraphs)

Tip

Start with your main spec, then add robustness checks one at a time. Commit after each working version.

After Spring Break: What's Coming

Week	Date	Focus
8	Mar 26	Main model estimation; robustness
9	Apr 2	Interpreting results; crafting the narrative
10	Apr 9	Writing: intro, background, empirical strategy
11	Apr 16	Figures, tables, slides; presentation prep
12	Apr 23	Peer review workshop; practice

Estimation → **interpretation** → **writing** → **presenting**.

Wrap-Up

1. Every design must answer the **five questions**
2. **Event studies** are the standard DiD diagnostic — learn `fixest::iplot`
3. Claude Code is a tool; **you are the economist**
4. **Preliminary results due March 27**

Have a great spring break. Come back with results.